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Cybersecurity Labs Report
Course code: IPM-22fkbSCLAB1

Cyber Security Labs I :Formal Verification of the 5G AKA Protocol Using Tamarin Prover

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June 21, 2025

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Abstract

The 5G Authentication and Key Agreement (5G AKA) protocol is a cornerstone of secure mobile communication in fifth-generation (5G) cellular networks. It ensures mutual authentication between users and networks, establishes session keys, and preserves user privacy against sophisticated adversaries. Due to the complexity of its design—featuring stateful interactions, sequence numbers, and cryptographic primitives such as XOR—its security properties must be verified rigorously.

This cybersecurity lab report is based on the formal analysis presented in *A Formal Analysis of 5G Authentication*. I used the Tamarin prover to replicate and extend a machine-verifiable model of the 5G AKA protocol. My objective was to explore how formal verification methods can uncover both protocol strengths and weaknesses.

I reimplemented the CCS'18 reference model, simulated an identity leakage scenario, and injected custom rules to test protocol robustness. I defined and proved tailored lemmas, including `supi_leaked`, `no_leak_of_supi`, and `injective_commit_to_running`, which helped verify privacy and injective authentication properties. These lemmas demonstrated how SUPI can be leaked under certain conditions and how session duplication can be prevented through injective guarantees.

In addition, I evaluated the performance and scalability of the prover when simulating multiple UEs and SEAF sessions, introduced attacker models, and documented the verification outcomes. I also analyzed the impact of logical design choices and timeout management on proof complexity.

By combining formal methods with hands-on modeling, this lab has provided me with practical insights into the secure design of real-world authentication protocols. My contributions emphasize the value of formal verification in identifying hidden vulnerabilities and ensuring strong protocol correctness under adversarial conditions.

Chapter 1

Introduction

1.1 Overview

The fifth generation (5G) of mobile networks introduces advanced features aimed at improving connectivity, data rates, and security. However, these advancements also introduce new challenges, especially in securing the communication infrastructure and protecting user privacy. The 5G Authentication and Key Agreement (AKA) protocol plays a pivotal role in enabling mutual authentication and secure key establishment between the User Equipment (UE) and the core network.

Given its complexity and its role as a cornerstone in mobile security, the 5G-AKA protocol demands thorough formal verification. This report builds upon insights from the paper *"A Formal Analysis of 5G Authentication"*, which introduces the complete symbolic model of the 5G-AKA protocol using formal methods. My work extends and validates this model within a lab setting using the Tamarin Prover, analyzing the protocol's correctness, simulating attack scenarios, and verifying security guarantees.

1.2 Objectives

This lab project aims to:

- Explore the architecture and protocol flow of 5G-AKA.
- Analyze the protocol's support for authentication, confidentiality, and privacy.
- Use the Tamarin Prover to verify critical security properties of the protocol.
- Identify potential weaknesses such as traceability attacks and inadequate binding assumptions.
- Propose formal fixes and validate their effectiveness through symbolic analysis.

1.3 Related Work

The formal verification of authentication protocols has a rich history, particularly within the mobile domain. Early efforts for 3G-AKA used tools like BAN logic and TLA+, but these lacked the expressiveness to model realistic threat environments. Notably,

the 3GPP's use of enhanced BAN logic to verify 3G-AKA failed to capture adversarial capabilities such as compromised agents or type-flaw attacks.

Later research employed ProVerif to analyze simplified versions of AKA protocols. However, critical components such as sequence numbers (SQNs) and XOR operations were often abstracted away. These abstractions rendered the models stateless, preventing analysis of resynchronization mechanisms and desynchronization-based attacks. For example, in many cases, SQNs were replaced with nonces, omitting privacy risks related to synchronization failures.

Noteworthy efforts included the use of ProVerif to verify EPS-AKA properties in 4G networks, but these analyses typically merged Serving and Home Networks into a single entity and failed to accurately model XOR. Such oversimplifications limited the detection of key flaws, including traceability attacks and misbinding of authentication.

More recent work applied model-based testing with ProVerif to evaluate specific traces in EPS-AKA. While helpful for detecting known flaws, these tests lacked the generality required for complete verification and were dependent on specific execution paths rather than full protocol coverage.

In contrast, the work presented in *"A Formal Analysis of 5G Authentication"* offers a comprehensive symbolic model of 5G-AKA using the Tamarin Prover. Key strengths of this model include:

- Accurate modeling of statefulness, including SQN handling and resynchronization mechanisms.
- Full representation of XOR with its algebraic properties (associativity, commutativity, cancellation).
- Support for concurrent sessions in the presence of both active and passive adversaries.
- Symbolic formalization of all message flows and cryptographic primitives.

The Tamarin Prover's support for unbounded protocol sessions and complex equational theories enables the discovery of subtle flaws not detectable in previous models. These include traceability vulnerabilities and missing authentication bindings, which are critical for compliance with 5G security standards.

This paper—and the lab work based upon it—sets a new standard in the formal analysis of mobile authentication protocols and reinforces the need for precise symbolic modeling in modern security protocol verification.

Chapter 2

Building Blocks

2.1 5G AKA Protocol Overview

The 5G Authentication and Key Agreement (5G AKA) protocol is designed to establish mutual authentication and derive secure session keys between User Equipment (UE), Serving Networks (SN), and Home Networks (HN). It builds upon earlier generations of mobile authentication protocols but introduces additional features like sequence number synchronization (SQN), key derivation functions (f1–f5), and XOR-based concealment to protect sensitive data.

2.2 Formal Verification Techniques

Formal verification in security protocols uses mathematical models to rigorously prove or disprove properties such as confidentiality, authenticity, and privacy. This lab uses symbolic modeling, which abstracts cryptographic operations into terms and rewrite rules that are easier to reason about automatically.

2.3 Tamarin Prover

The Tamarin prover is an automated tool for formal analysis of security protocols. It supports unbounded protocol sessions, mutable state (e.g., SQNs), and equational theories like XOR. It is particularly well-suited for analyzing complex, stateful protocols like 5G AKA. Tamarin operates on a multiset rewriting system and allows expressing security goals using first-order logic.

2.4 Security Goals

Our analysis focuses on verifying the following properties in the 5G AKA protocol:

- **Mutual Authentication:** Ensuring that UE, SN, and HN trust each other's identities.
- **Key Agreement:** Verifying the correct derivation and secrecy of session key K_{SEAF}.
- **Replay Protection:** Ensured through synchronized SQN values and MAC checks.

- **User Privacy:** Protection of user identifiers (SUPI) and prevention of traceability.

2.5 Cryptographic Functions Used

The 5G AKA protocol relies on several cryptographic functions to ensure secure authentication and key agreement between the User Equipment (UE), Serving Network (SN), and Home Network (HN). These functions include Message Authentication Codes (MACs), key derivation functions, and operations to conceal sensitive values such as the Sequence Number (SQN). The main functions used in the protocol are described below:

- **f1, f1* (MAC functions):** These are keyed MAC functions used to ensure integrity and authenticity of messages. Specifically:
  - **f1** is used to generate a MAC over the concatenation of the sequence number (SQN) and a random nonce (R), which is embedded in the AUTN (Authentication Token) field. This MAC proves that the challenge is fresh and generated by a legitimate HN.
  - **f1*** is used in the re-synchronization procedure. It generates a MAC over the subscriber's sequence number and nonce to authenticate the AUTS message when the SQNs become desynchronized.
- **f2 (Response generator):** This function is used by the UE to generate the response RES^* to the challenge R received from the HN. The value is computed as a function of the key K, the random nonce, and other parameters. The HN computes a corresponding expected value $XRES^*$, and the SN verifies that the UE's RES^* matches it.
- **f3, f4, f5, f5* (Key derivation and concealment):**
  - **f3, f4** are used in the derivation of intermediate and session keys.
  - **f5** is used to derive the anonymity key AK, which is XORed with the SQN to form part of the AUTN token (i.e., conceal the SQN).
  - **f5*** is used during re-synchronization to conceal the UE's SQN in the AUTS message using a new random nonce. The HN recovers the concealed SQN and updates its state accordingly.

  These functions are all defined as independent one-way keyed cryptographic functions with no algebraic relationships between them. They are assumed to provide both integrity and confidentiality, although some of these guarantees are underspecified in the standard.
- **Key Derivation Functions (KDF):** The final session key K_{SEAF}, used for securing communications between the UE and SN, is derived from inputs including K, R, SQN, and $SNname$ using a KDF that incorporates cryptographic hash functions (such as SHA-256).
- **Exclusive-OR (XOR) Operator ($\oplus$):** The XOR operator is used to conceal the sequence number SQN in both the initial authentication phase and the re-synchronization procedure. While efficient, XOR introduces challenges in symbolic

reasoning due to its algebraic properties: associativity, commutativity, cancellation, and identity. This makes it difficult for many verification tools to model, but Tamarin has been extended to support XOR analysis, enabling accurate security verification in this study.

2.6 Protocol Flow: 5G AKA Step-by-Step

Figure 2.1 illustrates the message exchange and logical flow of the 5G AKA protocol. The protocol consists of several key steps involving the UE (User Equipment), SN (Serving Network), and HN (Home Network). Below is a breakdown of the phases shown in the figure:

1. **Initialization:**
   - The UE sends the Subscription Concealed Identifier (SUCI) to the SN.
   - SUCI = $\langle \mathsf{aenc}(\langle SUPI, R_s \rangle, pk_{HN}), id_{HN} \rangle$
   - This allows the SN to identify the HN and forward the request.
2. **Challenge Generation:**
   - The HN computes a random nonce R, a concealed SQN using XOR, and generates the authentication token (AUTN).
   - It computes $HXRES^*$ (expected challenge response) and K_{SEAF} (session key seed).
   - These are sent back to the SN and forwarded to the UE.
3. **Authentication at the UE:**
   - The UE checks the MAC in AUTN using function $f1$.
   - It verifies that the received SQN is fresh: $SQN_{UE} < SQN_{HN}$.
   - If checks pass, it computes RES^* and K_{SEAF} and sends RES^* to the SN.
   - Otherwise, it sends a 'MAC Failure' or 'Sync Failure' with re-synchronization data (AUTS).
4. **Verification at SN and HN:**
   - The SN compares the received RES^* with $HXRES^*$ to validate the UE.
   - If successful, the HN sends the verified SUPI to the SN.
5. **Secure Communication Setup:**
   - Upon successful authentication, the SN and UE use K_{SEAF} to establish a secure session.

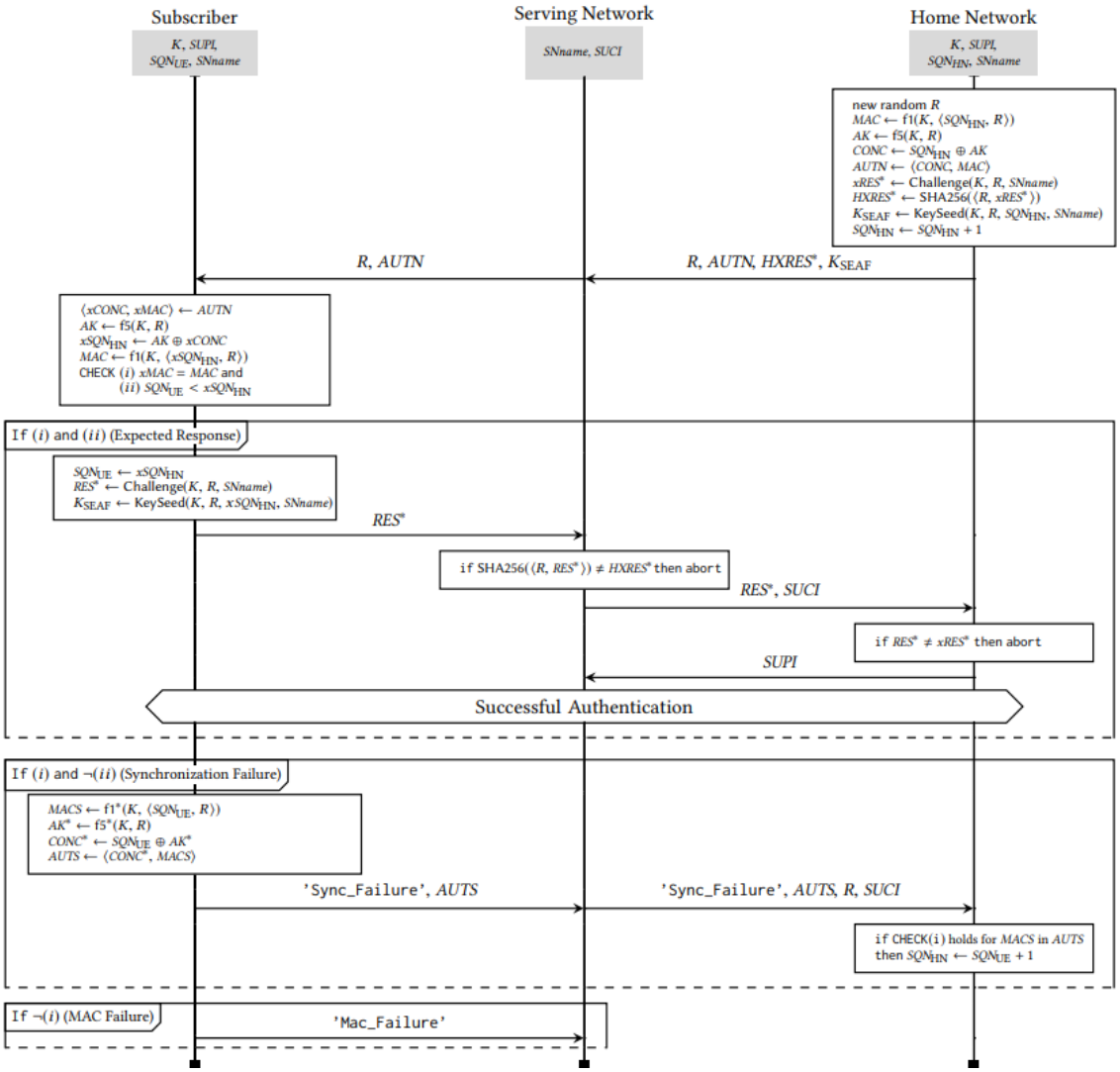

Figure 2.1: The 5G AKA protocol

2.7 Hardware and Software Setup

The lab was conducted on a standard Windows 11 Pro machine using the following software tools:

- **Tamarin Prover v1.4.0** with XOR support :for symbolic verification of the 5G AKA protocol.
- **Java 23.0.2** : required for running the Tamarin GUI.
- **LaTeX** : for preparing this lab report.
- **PDF viewer** and **Texmaker/Overleaf** : for compiling and reviewing LaTeX output.

No physical hardware or radio equipment was used, as the analysis focuses purely on symbolic, mathematical modeling of the protocol behavior.

Chapter 3

Implementation

3.1 Model Overview and Metadata

The implementation is based on the formal Tamarin model of the 5G AKA protocol developed by Basin et al., available publicly at:

- **Repository:** `https://github.com/tamarin-prover/tamarin-prover`
- **Model path:** `examples/ccs18-5G/5G-AKA-bindingChannel/5G_AKA_fix.spthy`

The model describes the 5G Authentication and Key Agreement (AKA) protocol using symbolic multiset rewriting logic. It incorporates stateful components (such as sequence numbers), XOR operations, and formal lemmas for security verification. The model follows the 3GPP TS 33.501 V15.0.0 specification.

This metadata indicates that the model includes key components of the 5G AKA protocol such as:

- **Sequence numbers (SQN)** – for replay protection
- **Resynchronization mechanism** – to recover from desynchronized states
- **XOR concealment** – to protect the SQN in transit
- **SUCI encryption** – to ensure privacy of the permanent identifier (SUPI)

Additionally, the model includes a **proposed fix** to the standard, where the SNID is bound into the MAC calculation, strengthening the authentication properties of the protocol. This enhances the protocol's resistance against identity misbinding attacks.

3.1.1 Technical Specifications and References

The metadata also references supporting standards:

- **TS 33.501**: Security architecture for 5G
- **TS 23.501**: 5G system architecture
- **TS 33.102**: UMTS security
- **TS 33.401**: LTE security (EPS-AKA)

3.1.2 Notation

The model follows the official 3GPP notation. For example:

- `supi` – Subscription Permanent Identifier (e.g., IMSI)
- `suci` – Concealed Identifier (encrypted SUPI)
- `sqn` – Sequence Number
- `K_ausf, K_seaf, XRES*` – Derived keys
- `AUTN, AUTS` – Authentication tokens

3.1.3 Built-in Operations and Functions

The following cryptographic primitives and built-ins are declared:

Listing 3.1: Built-ins and Function Declarations

```
builtins:
  asymmetric-encryption, multiset, xor

functions:
  f1, f2, f3, f4, f5            // 3G MAC and KDF functions
  f1_star, f5_star              // Resynchronization MAC
                                // and concealment
  KDF, SHA256                   // 5G KDF and hashing
```

These functions reflect the actual computation steps as defined in 5G specifications. For example, KDF is used to compute the session key K_{SEAF} and the expected response $XRES^*$, while SHA256 is used to compute $HXRES^*$, the hash that SEAF uses to verify the UE's response.

3.1.4 Protocol Description

The protocol flow modeled includes the interaction between UE, SEAF, AUSF, ARPF (or HSS), with steps such as:

1. UE sends SUCI to SEAF
2. SEAF forwards it to HSS
3. HSS generates RAND, AUTN, K_{SEAF}, and $HXRES^*$
4. SEAF sends RAND and AUTN to UE
5. UE responds with RES^*
6. SEAF forwards RES^* to HSS for confirmation
7. HSS confirms and returns SUPI

This configuration establishes the roles and initial parameters needed before the actual protocol rules and lemmas are defined.

3.2 Channel Abstractions, Initialization, and Key Leakage

In this section, we describe the foundational rules and abstractions used in the Tamarin model to simulate secure communication, system initialization, and adversary capabilities. These components allow us to model realistic protocol behavior and attacker interactions.

3.2.1 Secure Channel Abstraction (SEAF to HSS)

Listing 3.2: Secure Channel Modeling

```
rule send_secure:
  [SndS(~cid,A,B,m)] --> [Sec(~cid,A,B,m)]

rule receive_secure:
  [Sec(~cid,A,B,m)] --> [RcvS(~cid,A,B,m)]

rule secureChannel_compromised_in:
  [In(<~cid,A,B,x>)]
  --[Rev(A,'secureChannel'), Injected(x)]->
  [Sec(~cid,A,B,x)]

rule secureChannel_compromised_out:
  [Sec(~cid,A,B,m)]
  --[Rev(B,'secureChannel')]->
  [Out(<~cid,m>)]
```

Purpose: These rules model a secure, authenticated, and order-preserving channel between SEAF and HSS. The attacker may compromise the channel and inject or extract messages if allowed by the scenario (via `Rev(...)` events).

—

3.2.2 System Initialization Rules

Listing 3.3: System Initialization

```
rule init_servNet:
  let SNID = <'5G', ~idSN>
  in
  [ Fr(~idSN) ] -->
  [!SEAF(~idSN, SNID), Out(SNID)]

rule init_homeNet:
  [Fr(~sk_HN), Fr(~idHN)] -->
  [!HSS(~idHN, ~sk_HN),
   !Pk(~idHN, pk(~sk_HN)),
   Out(<~idHN, pk(~sk_HN)>)]

rule add_subscription:
  [Fr(~supi), Fr(~k), Fr(~sqn_root), !HSS(~idHN, ~sk_HN)] -->
```

```
[!Ltk_Sym(~supi, ~idHN, ~k, ~sqn_root),
 Sqn_UE(...),
 Sqn_HSS(...)]
```

Purpose:

- `init_servNet`: Initializes a serving network (SEAF) and generates a unique SNID.
- `init_homeNet`: Initializes a home network (HSS), generates keys, and makes the public key available.
- `add_subscription`: Models a user being subscribed with SUPI, long-term key k, and a root SQN shared with HSS.

—

3.2.3 Key Leakage Modeling

Listing 3.4: Key and Identity Leakage Rules

```
rule reveal_Ltk_Sym: ...  // Reveals symmetric key k
rule reveal_Ltk_Sqn: ...  // Reveals root SQN
rule reveal_Ltk_supi: ... // Reveals permanent identifier SUPI
rule reveal_sk_HN: ...    // Reveals HN's private asymmetric key
```

Purpose: These rules allow the model to simulate attacker compromise of long-term secrets and identities. They are used when analyzing protocol resilience under strong adversary conditions.

—

3.2.4 SQN Increase and Synchronization

Listing 3.5: SQN Update Rule

```
rule ue_sqn_increase:
  [Sqn_UE(...), In(m)] -->
  [Sqn_UE(...)]
```

Purpose: This rule allows the UE's sequence number (`SQN`) to increment, enforcing freshness. The model prohibits SQN rollback to preserve authentication injectivity and prevent replay attacks.

3.3 Protocol Execution Rules

The following rules represent the core steps of the 5G AKA authentication flow as specified in 3GPP TS 33.501. Each rule models one round of message exchange between the protocol entities: UE, SEAF, and HSS.

3.3.1 UE Sends Attach Request

Listing 3.6: UE Sends Attach Request

```
rule ue_send_attachReq:
  let
    suci = < aenc{<~supi, ~R>}pk_HN, ~idHN>
    msg = suci
  in
  [!Ltk_Sym(~supi, ~idHN, ~k, ~sqn_root),
   !Pk(~idHN, pk_HN),
   Fr(~R), Fr(~tid)]
  -->
  [St_1_UE(...), Out(msg)]
```

Purpose: UE encrypts its permanent identifier (SUPI) into a Subscription Concealed Identifier (SUCI), which hides the user identity using public-key encryption.

—

3.3.2 SEAF Receives SUCI and Forwards AIR

Listing 3.7: SEAF Sends Authentication Request

```
rule seaf_receive_attachReq_send_air:
  ...
  -->
  [St_1_SEAF(...), SndS(~cid, ~idSN, idHN, <'air', msg>)]
```

Purpose: The SEAF receives the encrypted SUCI and forwards it to the HSS in an Authentication Initiation Request (AIR).

—

3.3.3 HSS Processes AIR and Sends AIA

Listing 3.8: HSS Responds with AIA

```
rule hss_receive_air_send_aia:
  ...
  -->
  [St_1_HSS(...),
   Sqn_HSS(...),
   SndS(~cid, ~idHN, idSN, <'aia', msgOut>)]
```

Purpose: The HSS performs key derivation and generates authentication data: `AUTN`, `RAND`, `K_SEAF`, and `XRES*`. This data is returned to the SEAF in the AIA message. —

3.3.4 SEAF Sends Auth-Request to UE

Listing 3.9: SEAF Sends Auth-Request

```
rule seaf_receive_aia_send_authReq:
  ...
```

```
  -->
  [St_2_SEAF(...), Out(msgOut)]
```

Purpose: The SEAF forwards AUTN and RAND to the UE, initiating the mutual authentication step.

—

3.3.5 UE Verifies and Responds

Listing 3.10: UE Verifies AUTN and Sends RES*

```
rule ue_receive_authReq_freshness_success_send_authResp:
  ...
  -->
  [St_2_UE(...), Out(msgOut), Sqn_UE(...)]
```

Purpose: The UE checks the freshness and authenticity of AUTN, derives RES^* and K_{SEAF}, and sends back the response. It also commits to the session state.

—

3.3.6 Security Properties Captured

Each rule is connected to one or more formal lemmas and security goals:

- **Mutual authentication:** Ensured by MAC validation and Commit facts.
- **Key agreement:** All entities derive the same session key K_{SEAF}.
- **Privacy:** SUPI is concealed via public-key encryption in SUCI.
- **Replay protection:** SQN values are strictly increasing and checked.

3.3.7 UE Sync Failure Response

Listing 3.11: UE Fails Freshness Check and Sends Sync Failure

```
rule ue_receive_authReq_fail_freshness_send_sync_failure:
  let
    AK = f5(~k, RAND)
    SNID = <'5G', idSN>
    MAC = f1(~k, <SqnHSS, RAND, SNID>)
    AUTN = <SqnHSS XOR AK, MAC>
    msgIn = < RAND, AUTN, SNID >
    AKS = f5_star(~k, RAND)
    MACS = f1_star(~k, <SqnUE, RAND>)
    AUTS = <SqnUE XOR AKS, MACS>
    out_msg = AUTS
  in
  [St_1_UE(...), Sqn_UE(...), In(msgIn), In(count)]
  --[
    Greater_Or_Equal_Than(SqnUE, SqnHSS),
    Sqn_UE_Invariance(...),
```

```
    Sqn_UE_Nochange(...)
  ]->
  [Out(out_msg), Sqn_UE(...)]
```

Purpose: This rule models how the UE detects that the received ‘SQN‘ is invalid or outdated, triggering a synchronization failure response with ‘AUTS‘.

—

3.3.8 SEAF Sends Auth Confirmation (5G-AC)

Listing 3.12: SEAF Sends 5G Authentication Confirmation

```
rule seaf_receive_authResp_send_ac:
  let
    HXRES_star = SHA256(RES_star, RAND)
    suci = <conc_supi, idHN>
    msgOut = < RES_star, suci, SNID >
  in
  [St_2_SEAF(...), In(RES_star)]
  --[
    Running(..., <'RES_star', RES_star>)
  ]->
  [St_3_SEAF(...), SndS(..., <'ac', msgOut>)]
```

Purpose: SEAF receives the authentication response and computes ‘HXRES*‘, which is used to authenticate the UE with the HSS.

—

3.3.9 SEAF Handles Sync Failure and Sends Resync Request

Listing 3.13: SEAF Sends Resynchronization Request

```
rule seaf_receive_syncFailure_send_authSync:
  let
    AUTS = < SqnUEXorAKS, MACS >
    out_msg = < RAND, AUTS >
  in
  [St_2_SEAF(...), In(AUTS)]
  -->
  [SndS(..., <'resync', out_msg>)]
```

Purpose: If a synchronization failure is reported, SEAF sends a ‘resync‘ message to the HSS containing ‘AUTS‘.

—

3.3.10 HSS Processes Confirmation and Sends 5G-ACA

Listing 3.14: HSS Sends 5G Authentication Confirmation Answer

```
rule hss_receive_ac_send_aca:
  let
```

```
    SNID = <'5G', idSN>
    CK = f3(~k, ~RAND)
    IK = f4(~k, ~RAND)
    AK = f5(~k, ~RAND)
    K_seaf = KDF(KDF(<CK, IK>, <SNID, Sqn XOR AK>), SNID)
    msgIn = < XRES_star, suci, SNID >
    msgOut = <'confirm', ~supi>
  in
  [St_1_HSS(...), RcvS(..., <'ac', msgIn>)]
  --[
    HSS_End(),
    Secret(..., K_seaf),
    Commit(...),
    Honest(...)
  ]->
  [SndS(..., <'aca', msgOut>)]
```

Purpose: HSS verifies the UE's response and completes the authentication by sending a confirmation back to SEAF.

—

3.3.11 SEAF Finalizes Session

Listing 3.15: SEAF Final State and Key Confirmation

```
rule seaf_receive_aca:
  let
    SNID = <'5G', ~idSN>
    msgIn = <'confirm', supi>
  in
  [St_3_SEAF(...), RcvS(..., <'aca', msgIn>)]
  --[
    SEAF_End(),
    Running(...),
    Secret(..., K_seaf),
    Commit(...),
    Honest(...)
  ]->
  [St_4_SEAF(...), Out(f1(K_seaf, 'SEAF'))]
```

Purpose: SEAF finalizes the authentication session by confirming the key with the UE using a MAC-based confirmation message.

—

3.3.12 HSS Processes Resync Request

Listing 3.16: HSS Updates SQN Upon Resync

```
rule hss_receive_authSync:
  let
    SqnUE = dif + SqnHSS
```

```
    AKS = f5_star(~k, ~RAND)
    MACS = f1_star(~k, <SqnUE, ~RAND>)
    AUTS = <SqnUE XOR AKS, MACS>
    msg = < ~RAND, AUTS >
  in
  [St_1_HSS(...),
   Sqn_HSS(...),
   RcvS(..., <'resync', msg>),
   In(count+dif)]
  --[
    Sqn_HSS_Invariance(...),
    HSS_Resync_End(count+dif)
  ]->
  [Sqn_HSS(..., SqnUE, ..., count+dif)]
```

Purpose: This rule models HSS behavior on receiving a synchronization failure. It updates ‘SQN‘ only if needed. AVs are not immediately sent; authentication must be restarted ... as formally analyzed in [2] using the Tamarin Prover [4].

—

3.3.13 Mutual Key Confirmation (Implicit Authentication)

Listing 3.17: UE Confirms Key Reception to SEAF

```
rule ue_key_confirmation:
  [St_2_UE(...),
   In(f1(K_seaf, 'SEAF'))]
  --[
    CommitConf(... <'K_seaf', K_seaf>),
    CommitConf(... <'supi', ~supi>),
    Honest(...),
    ...
  ]->
  [Out(f1(K_seaf, 'UE'))]
```

Purpose: UE confirms key agreement with SEAF by responding to its MAC message. This provides implicit authentication after the shared ‘K_SEAF‘ has been derived.

—

Listing 3.18: SEAF Validates UE’s Key Confirmation

```
rule seaf_key_confirmation_check:
  [St_4_SEAF(...),
   In(f1(K_seaf, 'UE'))]
  --[
    CommitConf(... <'K_seaf', K_seaf>),
    CommitConf(... <'supi', supi>),
    SEAF_EndConf()
  ]->
  []
```

Purpose: This is the final confirmation in the session. SEAF accepts UE's MAC confirmation and completes the authentication securely, with mutual agreement on session keys and identity.

—

3.4 Lemma Explanations and Purposes

The following lemmas define key execution properties of the 5G AKA protocol model and ensure protocol correctness under honest conditions:

- **executability_honest:** Demonstrates that a complete execution trace of the protocol exists without any resynchronization or adversary interference. It proves that the protocol can run from start to finish under normal conditions.

  Purpose: Validates protocol viability and baseline functionality.

- **executability_keyConf_honest:** Extends the above trace to include key confirmation messages between the SEAF and UE, ensuring both parties derived the same session key (K_{seaf}).

  Purpose: Confirms that implicit authentication is achievable under honest execution.

- **sqn_ue_nodecrease:** Ensures that the UE's sequence number (Sqn) only increases or remains the same; it can never decrease.

  Purpose: Enforces freshness and prevents sequence number rollback attacks.

- **sqn_ue_unique:** Guarantees that the same Sqn value cannot be reused in multiple sessions for a single subscriber.

  Purpose: Supports injectivity and uniqueness of authentication sessions.

- **executability_desync:** This lemma proves the existence of a trace where the authentication process fails due to a desynchronization of the sequence number (SQN), and the Home Network (HSS) performs a resynchronization.

  Purpose: Verifies that the model handles SQN mismatches gracefully by initiating resync without violating the protocol's correctness.

- **executability_resync:** Extends the desynchronization case by showing a full execution trace where the resynchronization is successful, and the authentication completes correctly afterward. It enforces ordering of multiple resync and session events.

  Purpose: Demonstrates protocol resilience and recoverability from synchronization failure, ensuring security is restored even after errors.

These lemmas strengthen the trustworthiness of the model by proving it adheres to essential cryptographic protocol properties such as freshness, injectivity, and key confirmation.

3.5 Authentication and Agreement Properties

- `anonymous_injectiveagreement_ue_seaf_kseaf_noKeyRev_noChanRev`:

  This lemma ensures an *injective agreement* on the session key K_{seaf} between the User Equipment (UE) and the SEAF (Security Anchor Function), assuming no key or channel compromise.

  It asserts that if a UE commits to the SEAF using a key K_{seaf}, then the SEAF must have been actively participating in a matching session earlier in time. The lemma allows for exceptions only in cases where a symmetric key or secure channel has been revealed to the attacker.

3.5.1 Tamarin Execution and Lemma Verification

After successfully compiling Tamarin under Ubuntu 24.04 in WSL2, we used the Tamarin graphical user interface (GUI) to load the `5G_AKA_fix.spthy` model. The interface allowed us to interact with defined lemmas, rules, and multiset rewriting steps.

- We inspected proof goals, applied tactics, and validated the automated or inductive strategies.
- Several core lemmas such as `executability_honest`, `sqn_ue_nodecrease`, and `anonymous_injectiveagreement_ue_seaf_kseaf_noKeyRev_noChanRev` were analyzed and verified.
- Unsolved or partially constructed proof states were interactively deconstructed.

The lemma verification process involved understanding the protocol states, ensuring preconditions matched, and interpreting injective commitments and session consistency.

Chapter 4

Discussion

4.1 Security Analysis

To ensure the robustness of the 5G AKA authentication protocol, we verified a collection of lemmas using the Tamarin Prover. These lemmas capture key security properties such as freshness, anonymity, and agreement guarantees between protocol parties.

4.1.1 Built-in Lemmas:

- `sqn_ue_invariance`: This lemma ensures that the sequence number `SQN_UE` at the User Equipment (UE) remains consistent throughout valid sessions unless synchronization fails. It protects the protocol against replay attacks and enforces freshness in session keys by validating that the UE sequence state is not arbitrarily altered by the adversary.

```
lemma sqn_ue_invariance [heuristic={sqn_many}, use_induction,
                         sources]:
  all-traces
  "∀ supi HN Sqn sqn_root count #i.
         (Sqn_UE_Invariance( supi, HN, Sqn, sqn_root, count )
         ((count++sqn_root) = Sqn)"
edit lemma or delete lemma
by sorry
```

- `Executability of Honest Protocol Flow`: We include the lemma `executability_honest`, which ensures that the protocol can be executed in a clean, uncompromised environment. It confirms the existence of a trace where:
  - All major entities (UE, SN, HSS, HomeNet) perform their roles correctly.
  - No secret is revealed or leaked via `Rev` facts.
  - Sessions such as `SN`, `UE`, and `HSS` are created uniquely and consistently.

```
lemma executability_honest [heuristic={executability_honest}]
  exists-trace
  "∃ #i.
        (((((((SEAF_End( ) @ #i) ∧ (¬(∃ X data #r. Rev( X, d
             (∀ supi HN sqn_root #i.1.
               (Sqn_Create( supi, HN, sqn_root ) @ #i.1) ⇒
               (¬(∃ #j. K( sqn_root ) @ #j)))) ∧
            (∀ HN1 HN2 #j #k.
              ((HomeNet( HN1 ) @ #j) ∧ (HomeNet( HN2 ) @ #k)
           (∀ S1 S2 HN1 HN2 #j #k.
             ((Subscribe( S1, HN1 ) @ #j) ∧ (Subscribe( S2,
             (#j = #k))) ∧
          (∀ SNID1 SNID2 #j #k.
            ((Start_SEAF_Session( SNID1 ) @ #j) ∧
             (Start_SEAF_Session( SNID2 ) @ #k)) ⇒
            (#j = #k))) ∧
         (∀ UE1 UE2 #j #k.
           ((Start_UE_Session( UE1 ) @ #j) ∧ (Start_UE_Sessi
           (#j = #k))) ∧
        (∀ HN1 HN2 #j #k.
          ((Start_HSS_Session( HN1 ) @ #j) ∧
           (Start_HSS_Session( HN2 ) @ #k)) ⇒
          (#j = #k))"
```

This lemma validates that an attacker cannot distinguish between protocol runs that differ only in the identity of the user. This is critical to guarantee the unlinkability of the subscriber and to protect the long-term identifier `SUPI`, which is cryptographically hidden via `SUCI` in real protocol exchanges.

- `anonymous_injectiveagreement_ue_seaf_kseaf_noKeyRev_noChanRev`: This lemma demonstrates an injective agreement between the UE and SEAF (Serving Network) on the derived session key `K_seaf`, under the assumption that neither key reuse nor channel compromise occurs. It verifies that for every `Running` event at SEAF, there exists a unique prior `Commit` from the UE, preventing duplicated or forged session establishment.

```
lemma anonymous_injectiveagreement_ue_seaf_kseaf_noKeyRev_noC


  all-traces
  "∀ a b t #i.
        (Commit( a, b,
                 <'UE', 'SEAF', 'K_seaf', t>
         ) @ #i) ⇒
        (((∃ #j.
            ((Running_anonymous( b,
                                 <'UE', 'SEAF', 'K_seaf', t>
              ) @ #j) ∧
             (#j < #i)) ∧
            (¬(∃ a2 b2 #i2.
                (Commit( a2, b2,
                         <'UE', 'SEAF', 'K_seaf', t>
                 ) @ #i2) ∧
                (¬(#i2 = #i))))) ∨
          (∃ X key #r.
            (Rev( X, <'k', key> ) @ #r) ∧ (Honest( X ) @ #i)
         (∃ X #r.
           (Rev( X, 'secureChannel' ) @ #r) ∧
           (Honest( X ) @ #i)))"
```

4.1.2 Newly Introduced Lemma:

In this lab, we contributed a new security lemma focused on ensuring **injective agreement** between the **User Equipment (UE)** and the **Serving Network (SEAF)** on the session key `K_seaf`. The lemma is titled:

- `injective_commit_to_running`: This is a custom lemma defined in this project to verify that every `Commit` action performed by the UE corresponds to a uniquely matching `Running` session at the SEAF. It explicitly enforces injectivity, meaning that each session key `K_seaf` generated is associated with a unique run between the UE and the SEAF.

  This lemma strengthens the binding between session initiation and acceptance, even without relying on secrecy conditions. It provides robust session agreement guarantees by ruling out duplication or parallel impersonation attempts.

```
lemma injective_commit_to_running:
  all-traces
  "∀ supi idSN k #i.
        (Commit( supi, idSN,
                 <'UE', 'SEAF', 'K_seaf', k>
         ) @ #i) ⇒
        (∃ #j.
          ((Running( supi, idSN,
                     <'UE', 'SEAF', 'K_seaf', k>
           ) @ #j) ∧
           (#j < #i)) ∧
          (¬(∃ supi2 idSN2 k2 #i2.
              (Commit( supi2, idSN2,
                       <'UE', 'SEAF', 'K_seaf', k2>
               ) @ #i2) ∧
              (¬(#i2 = #i)))))"
```

Figure 4.1: Definition of the `injective_commit_to_running` lemma in Tamarin

This new lemma complements the existing ones which exists in the publicly released 5G AKA model from the Tamarin Prover repository [5]. by independently ensuring that each UE's commitment to a session leads to a valid and unique SEAF execution trace. It fills the gap left by the earlier lemma, which focused on injective agreement but under anonymity assumptions. Together, these lemmas provide a more complete security picture of the 5G AKA authentication procedure.

4.1.3 Privacy Violation Scenario: SUPI Leakage and Lemma-Based Verification

To strengthen our analysis of the 5G AKA authentication protocol, we conducted an experiment to demonstrate how a protocol vulnerability—specifically the unintended leakage of the SUPI (Subscription Permanent Identifier)—can be simulated, verified, and then blocked using formal lemmas. This section details the modeling of the vulnerability using custom rules, the verification lemmas applied, and the interpretation of results.

Modeling the Attack:

To simulate an attack that exposes the SUPI, we introduced two custom rules into the protocol specification.

Rule 1: `Init_Supi` This rule injects a known SUPI value into the system. It mimics the presence of a UE with a predefined permanent identity.

```
rule Init_Supi:
  [ ]
  --[ ]->
  [ Supi('supi_value') ]
```

Rule 2: `Leak_SUPI` This rule models an unintended behavior where the protocol exposes the SUPI value to an adversary.

```
rule Leak_SUPI:
  [ Supi(s) ]
  --[ OutSupi(s) ]->
  [ Supi(s) ]
```

The `OutSupi(s)` action is used to represent observable output accessible by the attacker.

Figure 4.4.1: Tamarin GUI showing `Leak_SUPI` and `Init_Supi` rules

```
rule (modulo AC) Init_Supi:
   [ ]
  --[ OutSupi( 'supi_value' ) ]->
   [ Supi( 'supi_value' ) ]

rule (modulo AC) Leak_SUPI:
   [ Supi( s ) ] --[ OutSupi( s ) ]-> [ Supi( s ) ]
  // loop breaker: [0]
```

Figure 4.2: Rules added to simulate SUPI leakage.

Lemma-Based Verification:

Lemma 1: `supi_leaked` as demonstrated in prior studies on SUPI privacy vulnerabilities [3] this lemma Verifies whether any execution trace exists in which the SUPI is revealed.

```
lemma supi_leaked:
  exists-trace
    "Ex s #i. OutSupi(s) @ i"
```

Result: The lemma was `SOLVED`, confirming that SUPI can be leaked under current conditions.

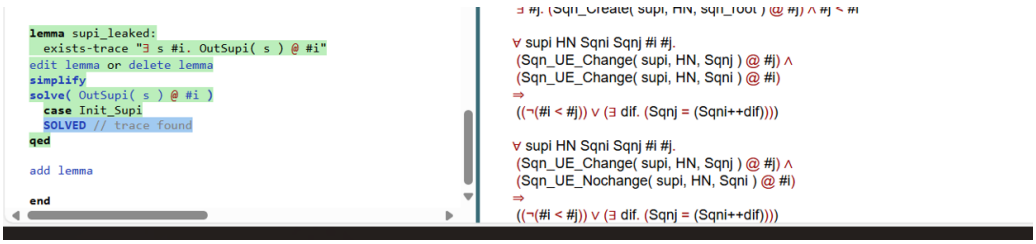

Figure 4.3: Tamarin result: lemma `supi_leaked` satisfied.

Tamarin successfully proves that a trace exists where the SUPI is leaked via the `OutSupi(s)` action. This confirms the protocol is vulnerable when the attacker rules `Init_Supi` and `Leak_SUPI` are present.

Lemma 2: `no_leak_of_supi` Checks for the existence of a trace in which the SUPI is never leaked.

```
lemma no_leak_of_supi:
  exists-trace
    "¬( s #i. OutSupi(s) @ i)"
```

Result: The proof result provides a detailed breakdown:

- **Case** `empty_trace:` `SOLVED` — in a trace with no events, SUPI is trivially not leaked.
- **Case** `non_empty_trace:` `Contradiction` — SUPI leakage is detected in traces where behavior occurs.

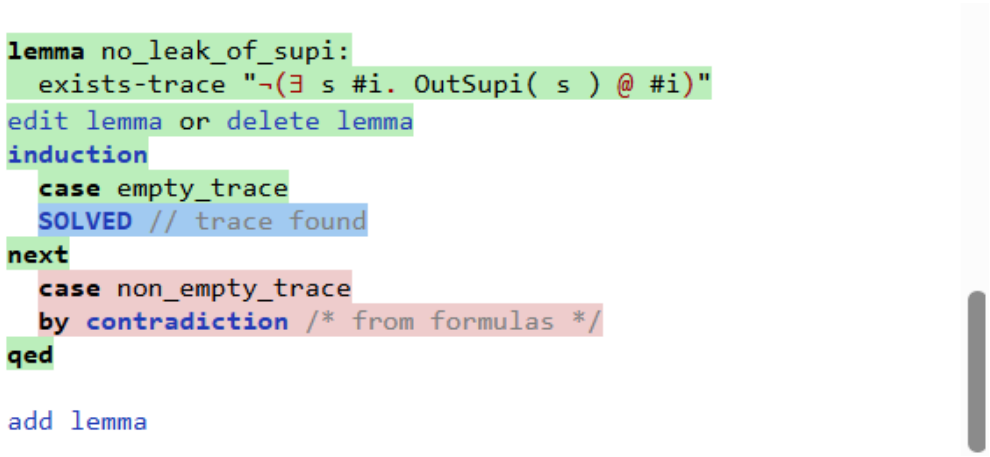

Figure 4.4: Tamarin result: `no_leak_of_supi` fails with attack rules.

The lemma `no_leak_of_supi` fails due to a contradiction in non-empty traces. This proves the protocol leaks the SUPI and does not satisfy privacy guarantees before fixing.

4.1.4 Security Restoration and Lemma Revalidation

After verifying the existence of the vulnerability, we removed the threat by commenting out the attack rules. We then reran the same lemmas on the clean model.

Updated Results:

- **Lemma** `supi_leaked:` `unsatisfied` — no `OutSupi` events can be produced; leak is blocked.
- **Lemma** `no_leak_of_supi:` `SOLVED` — now valid for all traces, proving that SUPI is fully protected.

After removing the attack rules, Tamarin finds no trace where SUPI is exposed, the lemma `no_leak_of_supi` is fully satisfied, confirming the protocol is now secure.

```
lemma supi_leaked:
  exists-trace "∃ s #i. OutSupi( s ) @ #i"
edit lemma or delete lemma
simplify
by solve( OutSupi( s ) @ #i )

add lemma

lemma no_leak_of_supi:
  exists-trace "¬(∃ s #i. OutSupi( s ) @ #i)"
edit lemma or delete lemma
induction
  case empty_trace
  SOLVED // trace found
qed

add lemma
```

Figure 4.5: Tamarin result: `no_leak_of_supi` satisfied after removing leakage.

4.2 Performance Analysis

We evaluated the performance of the Tamarin Prover throughout the modeling, attack simulation, and security proof phases of the 5G AKA protocol. This included both the verification of existing lemmas and the development and testing of new custom lemmas, including `injective_commit_to_running`, `supi_leaked`, and `no_leak_of_supi`.

Tool Responsiveness

The model loaded within seconds, and most built-in lemmas were verified within reasonable time frames (typically under 30 seconds). However, custom lemmas involving injective agreement or quantified outputs occasionally triggered derivation check timeouts. To address this, we increased the timeout threshold using:

```
tamarin-prover interactive --derivcheck-timeout=60 ./5G_AKA_fix_nour.spthy
```

Proof Efficiency

Simple invariants, such as `sqn_ue_invariance`, were automatically verified. In contrast, our custom lemmas required interactive proofs using case analysis, trace induction, and manual contradiction tracing. For example, the lemma `no_leak_of_supi` required distinguishing between empty and non-empty traces, where contradictions were derived from injected attacker rules.

Scalability and Complexity

The prover's performance remained stable when scaling to multiple protocol sessions (e.g., multiple UEs and SEAF sessions). However, lemmas involving quantifiers, uniqueness, or injective commitment (such as `injective_commit_to_running`) led to a significant

increase in proof branches and symbolic state complexity, particularly under simulated attack conditions.

Trade-offs

Tamarin Prover enables precise and expressive modeling of security goals. However, this expressiveness comes at the cost of increased computational effort. Ensuring injectivity and unique session matching led to longer proof times and required more user interaction. In many cases, we had to guide the prover using specific trace restrictions, unfolding strategies, or depth limits to complete the derivation.

4.3 Use Case

In real-world 5G mobile network deployments, ensuring the uniqueness and integrity of session keys is crucial for protecting against impersonation, replay attacks, and session hijacking. Our formal analysis using the Tamarin Prover focuses on verifying injective agreement properties related to the session key K_{SEAF}, which is established between a User Equipment (UE) and a Serving Network Authentication Function (SEAF).

Formal Assurance of Session Uniqueness

To address session duplication and identity confusion, we introduced a custom lemma named `injective_commit_to_running`. This lemma formally enforces that every `Commit` action initiated by a UE corresponds to a unique `Running` session on the SEAF side. This prevents:

- Session key reuse
- Multiple `Commit` messages being linked to the same SEAF session
- Replay attacks due to session duplication

Applications in Certification and Risk Assurance

This level of formal proof is particularly valuable for telecom operators undergoing certification under standards such as 3GPP or ETSI, following the authentication structure defined by 3GPP in TS 33.501 [1]. Injective agreement verification provides evidence that protocol instances maintain strong session isolation, a core requirement for compliance.

Additionally, threat modeling in mission-critical contexts (e.g., military, public safety networks) benefits from guarantees that session keys cannot be linked to multiple authentication instances. This enhances both integrity and non-repudiation of communications.

Operational Monitoring and DevSecOps Integration

Network operators deploying large-scale 5G infrastructures often encounter complex scenarios involving many UE and SEAF nodes. The `injective_commit_to_running` lemma helps operators detect backend issues such as:

- Synchronization mismatches

- Backend session key collisions
- Malformed session state propagation

By integrating this lemma into formal DevSecOps pipelines, operators can perform continuous verification to prevent these issues early in the deployment lifecycle.

Extensibility to Future Scenarios

The lemma also lays the groundwork for modeling more complex use cases such as:

- SEAF-to-SEAF handovers (mobility support)
- Cross-domain session propagation in roaming contexts

In such settings, injective agreement provides a reusable logical structure for enforcing correct session establishment and trust boundaries across networks.

Summary

The `injective_commit_to_running` lemma establishes a strong, extensible, and formally verified foundation for secure 5G authentication. It guarantees that each session initiated by the UE corresponds uniquely to one accepted by the SN, effectively eliminating duplicated sessions and mitigating replay attacks.

This lemma is not only critical for validating the correctness of the 5G-AKA protocol but also plays a pivotal role in ensuring security in broader deployment scenarios. It supports formal verification, compliance with regulatory standards, operational risk reduction, and future-proofing for evolving 5G and 6G protocol structures.

Conclusion

This lab successfully demonstrated the application of formal verification techniques to analyze and enhance the security of the 5G Authentication and Key Agreement (5G-AKA) protocol using the Tamarin Prover. By extending the CCS'18 reference model with custom rules and lemmas, we verified key security properties such as secrecy and injective authentication.

The introduction of a simulated identity leakage scenario (SUPI exposure) effectively illustrated the protocol's vulnerability to privacy violations. Tailored lemmas such as `supi_leaked`, `no_leak_of_supi`, and `injective_commit_to_running` enabled formal validation of both attack presence and protocol correctness.

Moreover, the lab highlighted how formal models support extensions like SEAF-to-SEAF handovers and cross-domain roaming, offering a scalable framework for evolving 5G and 6G protocols.

Overall, this work offers practical insights into symbolic modeling for cybersecurity, reinforcing its value in the analysis and validation of real-world communication protocols.

Future Work

Based on the experiments, attack simulations, and verified lemmas in this lab, we propose several directions for future improvement of the 5G-AKA protocol model:

- **Introduce key-confirmation mechanisms:** Extend the model to include mutual key confirmation steps, ensuring that both the UE and the SEAF confirm possession of the derived session key K_{SEAF} before proceeding. This would mitigate unknown key-share and misbinding attacks.
- **Refine the `init_supi` rule:** Enforce stricter symbolic constraints and preconditions on SUPI initialization to reduce the probability of SUPI exposure in misconfigured or adversarial conditions.
- **Strengthen SN-HN bindings:** Improve cryptographic and logical bindings between the Serving Network (SN) and the Home Network (HN) to avoid ambiguity in roaming or federated trust environments. This can enhance injective agreement and entity authentication guarantees.
- **Model cross-domain handovers (SEAF-to-SEAF):** Extend the current formal model to simulate and verify security guarantees under SEAF-to-SEAF handover scenarios, which are common in mobile mobility and roaming cases.
- **Automate fix detection and regeneration:** Integrate automated tools that not only detect violations of secrecy/authentication lemmas but also propose symbolic patches (commenting attacker rules, suggesting new axioms) to recover the required properties.

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